

Longitudinal Sentiment Analysis of Hallucinogen Use in Online Communities

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Abstract

Background

While the use of hallucinogenic substances is prevalent, research is often restricted in a traditional clinical setting due to legal regulations. It is necessary that researchers and policymakers have a robust understanding of the user experiences to make informed decisions related to substance use and harm reduction.

Objective

This study has three primary objectives:

1. To evaluate temporal changes in average sentiment of select substances.
2. To identify the distribution of emotions expressed in conversations related to specific substances.
3. To develop an interactive, publicly available web application in which users can easily view and interpret our results, allowing for comprehensive comparison and analysis of substances.

Methods

We conducted a longitudinal sentiment analysis of Reddit posts across nineteen substance terms. Text data were preprocessed to remove noise, including normalizing Unicode characters, lowercasing text, and removing Markdown formatting, HTML, and URLs. Cleaned text was analyzed using two transformer-based models: GoEmotions for discrete emotion classification and the Ki-Element Multilingual Valence-Arousal (VA) model for continuous valence scoring. Models were selected for their robustness to informal language and contextual understanding. Processing was performed in batches with multithreading to ensure computational efficiency. Sentiment outputs were aggregated at the substance levels to enable temporal and comparative analyses.

Results

We conducted a longitudinal sentiment analysis to examine patterns in emotional valence across users discussing each of the nineteen substances in our corpus, aligning posts relative to each user's first observed activity. In addition to these user-level patterns, we identified the primary emotion expressed in each post related to these substances and calculated the proportion of posts associated with each emotion for each substance, enabling comparison across substances. We integrated these results into a publicly available interactive Streamlit application.

Conclusions

Online discussions of hallucinogenic substances on platforms like Reddit provide valuable insight into user perceptions and behaviors. Our longitudinal analyses show that sentiment and emotion patterns can be meaningfully extracted from large-scale social media data, even for sensitive topics like illicit substance use. While overall sentiment shifts were modest, our approach identifies trends among frequent users and highlights common emotional responses. Approval was the most prevalent, followed by curiosity, amusement, confusion, and annoyance. These findings support the potential of scalable methods to inform harm reduction research and improve understanding of substance-related discourse.

1. Introduction

1.1 Background

The unsupervised, non-clinical use of hallucinogenic substances is prevalent. Despite this, empirical research on these substances remains constrained by regulatory barriers, ethical limitations, and controlled clinical settings that may not accurately reflect real-world use patterns. As a result, much of the existing literature captures structured therapeutic or laboratory settings rather than naturalistic, unsupervised consumption.

In parallel, online communities have emerged as spaces where individuals openly discuss their experiences with hallucinogens. Platforms such as Reddit provide large-scale, user-generated narratives that document subjective effects, dosing practices, harm reduction strategies, and long-term outcomes. These discussions represent a valuable, though complex, source of real-world evidence.

Understanding user-reported experiences is critical for researchers, clinicians, and policymakers seeking to design evidence-informed interventions, improve harm reduction messaging, and anticipate emerging substance trends. By systematically analyzing large-scale online discourse, we can uncover patterns of substance-effect associations and track evolving public sentiment in near real time.

1.2 Objectives

Understanding the ways drug users feel about the substances they take, and how these feelings change over time with prolonged use, is vital to comprehensively understand and manage substance use. To support this understanding, this study has three primary objectives:

1. To evaluate temporal changes in average sentiment of select substances.
2. To identify the distribution of emotions expressed in conversations related to specific substances.
3. To develop an interactive, publicly available web application in which users can easily view and interpret our results, allowing for comprehensive comparison and analysis of substances.

2. Methods

2.1 Data & Preprocessing

Our initial corpus of data included over twenty-nine million posts and comments collected from sixteen unique subreddits between 2008 and 2022 (Appendix A). These data consist of unstructured text conversations in which users primarily discuss substances of interest, including topics such as dosage, strain or variety, first-hand experiences, and recommendations.

For the purposes of this study, we restricted our analysis to comments rather than posts, as comments were found to be more informative and more directly aligned with our objectives. The resulting dataset includes 23,867,665 comments.

The dataset comprises comments from a range of Reddit communities with differing scopes. Some subreddits (e.g., r/LSD and r/MDMA) focus on discussion of a single substance, while others (e.g., r/TherapeuticKetamine and r/microdosing) center on specific use cases or patterns of consumption. Additional communities support broader discussions spanning multiple substances and contexts.

Comments from all included subreddits were aggregated into a single corpus, which was used for subsequent preprocessing and modeling.

2.1.1 Preprocessing

Free-text social media data presents several challenges during cleaning and preparation for analysis. Unlike structured clinical or survey datasets, user-generated text frequently includes unconventional punctuation, fragmented sentence structure, slang, abbreviations, emojis, hyperlinks, markdown formatting, and spelling errors. In addition, Reddit comments may contain automated bot responses or repeated scripted language that does not reflect an authentic user experience. Careful preprocessing was necessary to reduce noise while preserving content relevant to user-reported experiences. This preprocessing includes basic cleaning steps including bot comment and hyperlink removal, as well as text normalization. The preprocessed data was used for both the substance-effect and longitudinal sentiment analyses.

2.1.2 Cleaning and Assumptions

To prepare the corpus for analysis, we implemented a structured cleaning pipeline guided by the context of hallucinogen-related discussions and the anticipated requirements of downstream modeling.

We first identified likely bot-generated comments. This was accomplished by flagging known bot usernames, particularly accounts publicly labeled as automated moderators or information bots, and by detecting highly repetitive language characteristic of automated responses. Comments identified as bot-generated were removed from the corpus to ensure that substance-effect associations reflected organic user experiences rather than automated moderation notices or scripted informational posts.

We then removed URLs and embedded hyperlinks from the text. Links typically redirect to external resources such as research articles, commercial vendors, or media content and do not directly contribute to the experiential narratives under study. Removing URLs reduced extraneous noise and improved the interpretability of the remaining content. Figure 1 shows the number of comments present in our data before and after these cleaning procedures.

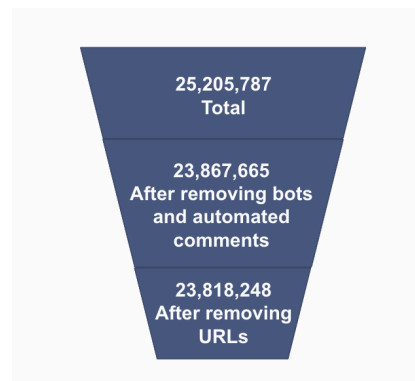


Figure 1. Count of Comments Remaining after Bot Comment and URL Removal.

Our original corpus contained 25,205,787 comments. We successfully identified and removed 1,338,122 bot- and automatically generated- comments. We further removed 49,417 comments containing URLs and embedded hyperlinks. The data used in the subsequent analyses contains 23,818,248 comments.

2.1.3 Text Normalization

After removing noisy comments, we normalized unicode, removed markdown formatting, and converted all text to lowercase. These steps were conducted to ensure standard formatting across comments, which improved the accuracy of the named-entity recognition model. Misspellings and equivalent names of substances were handled after the development of the named-entity recognition model, and will be discussed further in Section 2.3. At this stage, we retained punctuation formatting for the sentiment analysis model.

We opted to keep emojis and emoticons present in the final text data because of their significance in how users communicate about these substances. We found that certain substances can be referred to using emojis. For example, the horse emoji is frequently used to refer to ketamine, and the mushroom emoji is frequently used to refer to psilocybin. Removing these would result in the loss of those users' reported experiences.

2.1.4 Assumptions

In addition to preprocessing decisions, our analysis relies on several key assumptions about the content of the comments. These assumptions are considered reasonable given the context in which the data were generated:

- Most comments reflect first-hand experiences rather than second-hand or impersonal accounts.
- The majority of comments provide truthful and accurate descriptions of user experiences.
- Comments within substance-specific subreddits primarily pertain to that substance, and comments within use-case-specific subreddits primarily relate to the corresponding use case, particularly for the purposes of sentiment analysis.
- Frequency of comments is correlated with frequency of substance use. Therefore, users who discuss these substances more often are more likely to use them more frequently. In addition, sustained participation over time may serve as a proxy for prolonged exposure or continued use.

While comprehensive validation of these assumptions was beyond the scope of this study due to time and resource constraints, our experience working with the data, along with model outputs, suggests they are reasonable. Moreover, these assumptions are consistent with prior studies that utilize similar data sources for related research objectives.

2.2 Deduplication of Entities

After data cleaning and preprocessing, we first deduplicated substance entities. This was necessary because of the nature of our social media text data. Reddit users often use multiple different terms to refer to the same substance, whether those are street names, abbreviations, or simply misspellings. Without deduplicating substances before modeling, many substances would be repeated or missed. Deduplication of these substances consolidates the substance entities, ensuring no duplicate or split data.

The deduplication process was nontrivial, as there were over 25.2 million comments across all the subreddits in this project. Across those comments, users discussed thousands of substances with many different names, spellings, and slang terms. Therefore, an automated approach utilizing text analysis techniques was necessary to identify and collapse semantically similar, synonymous terms during the deduplication process. To achieve this, we:

- Trained Word2Vec models to generate vector embeddings of the words in our data.
- Developed initial lists of seed terms for substances and effects.

- Used the Word2Vec embeddings to identify the most semantically similar terms in our data to the seed terms.
- Added all of the terms above a certain similarity threshold (as determined by cosine distance) to the term lists.

This yielded expanded seed term lists that captured many terms we would not have originally thought to include. We then used these expanded term lists to extract all comments from the data that contained at least one term from the substance list. We then trained a named entity recognition (NER) model to identify substance entities using these comments. We did this by labeling substances across thousands of comments, and then feeding the resulting annotations to an NER model to train on. This resulted in a purpose-built NER model exclusively built to extract substances. We then ran this model on the entire dataset to extract all substances across all comments, generated fast text embeddings for each term, and finally clustered all substances and effects separately using HDBSCAN.

Terms that fell into the same cluster because of the similarity of their embedding vectors were manually reviewed to ensure they all referred to the same substance. Once reviewed, a canonical label was assigned to each cluster, after which all instances of the terms that made up the cluster across all comments in the data were replaced with the canonical label. After deduplication, we trained Word2Vec models.

2.1.1 Training Word2Vec Models

We trained Word2Vec models using the Gensim library to capture contextual relationships between terms in the corpus. All models were implemented using the skip-gram algorithm with an embedding dimension of 100. Each term is represented by a 100-dimensional embedding vector, where similarity in the vector space reflects semantic relatedness learned during training.

Three separate models were developed to account for differences in tokenization strategy. One model used spaCy tokenization, another used SciSpaCy tokenization, and a third extended the SciSpaCy pipeline to include bigram detection. This approach allowed the embeddings to better capture chemical compounds, preserve general slang terms, and represent multi-word expressions such as “magic mushrooms.” The resulting embeddings were used to expand seed terms by identifying semantically similar words in the embedding space and compiling them into keyword lists for querying relevant comments in the corpus.

2.1.2 Expanding Seed Terms

Seed term expansion was conducted through an iterative process applied separately to substance-related terms. Initial seed lists were constructed based on domain knowledge and prior review of the corpus. For each seed term, semantically similar terms were identified using cosine similarity in the learned embedding space.

A similarity threshold was selected for each term based on the number of retrieved neighbors, balancing coverage and precision. Threshold selection was guided by inspecting the trade-off between cosine similarity and the number of unique terms returned, allowing us to avoid overly broad expansions while retaining meaningful related terminology.

The expansion process was repeated iteratively, with newly accepted terms incorporated into the candidate pool. This procedure continued until a sufficiently comprehensive keyword list was generated to query relevant comments for annotation in support of the NER model. Figure 2 shows the seed term expansion charts for beacocystin.

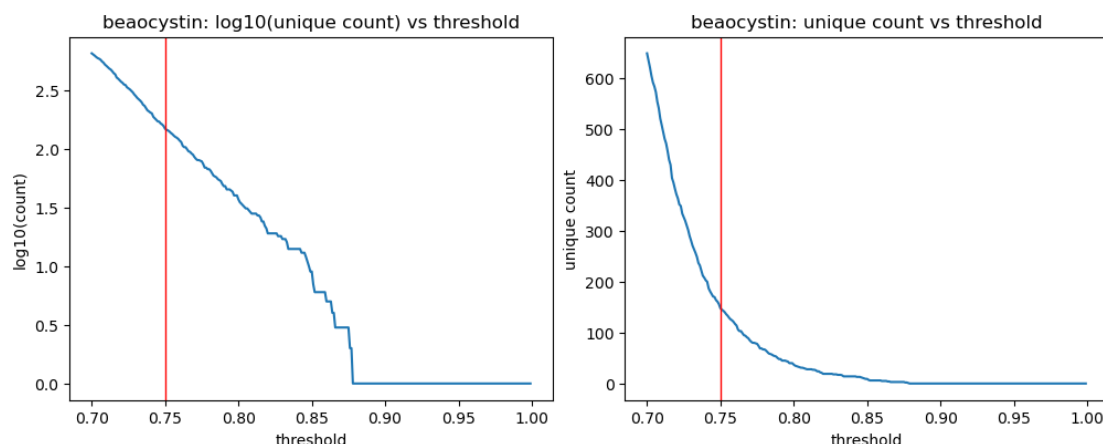


Figure 2: Seed Term Expansion Charts for the Substance Beacocystin

2.1.3 Training Named-Entity Recognition Model

After querying to select all comments in our data containing any of the words in the substance seed term list, we began the process of training our own custom NER model. We chose to train our own NER model for this step rather than using an off-the-shelf pretrained model because we wanted a model that would specifically and exclusively identify substances, and do so in our specific context of Reddit comments that have their own style and similar context.

Before we could train our NER model, we needed lots of annotated training data. We first generated this annotated data manually. We took a random sample of 1,000 comments and annotated them using Doccano, an open-source data labeling tool. We then read each of the 1,000 comments as they were presented in Doccano, identified any substances mentioned and labeled them accordingly. This included labeling slang terms and misspellings, as we wanted the NER model to identify those as well, so that they could all be collapsed together in the deduplication step.

2.1.4 LLM Annotation

Given the immense time required to manually annotate such a large volume of Reddit comments and the NER model's strong performance with more training data, we next decided to use an LLM to assist with the data annotation process. We used Gemini via the Gemini API, and prompted it using the prompt in Appendix B to label substances and effects in the Reddit comments and output a JSONL file in the same format as Doccano outputs so that we could plug it into our existing workflow. Using an LLM for annotation, we were able to annotate 10,000 comments in just 5 hours, which was significantly faster than manual annotation.

Before including the LLM-annotated comments, we trained our first custom NER model on the 1000 manually annotated comments. This model (spaCy-CNN) was trained from scratch using spaCy's configurable training pipeline. The model consists of a tok2vec feature-extraction layer using hash embeddings and a maxout window encoder, followed by a transition-based NER parser. The model was trained using Adam optimization on our annotated corpus without pretrained word vectors.

Once we had the 10,000 LLM-annotated comments, we made the decision to switch from the spaCy-CNN model to a fine-tuned transformer model (BERT-NER). The updated pipeline uses bert-base-

uncased as a contextual backbone via spaCy's TransformerListener interface, with the same transition-based NER decoder retrained on top of the transformer's subword representations. This shift moves from locally-windowed, non-contextual features to full bidirectional sentence-level context, which is better suited to the slang, abbreviations, and variable syntax present in the corpus. While BERT's pretraining reduces its dependence on labeled data relative to a from-scratch model, 1,000 annotations are marginal for reliably training the transition-based NER head; the expanded annotated data of 10,000 LLM-annotated comments plus the 1,000 human-annotated comments moves us into a range where the decoder can learn stable, consistent entity boundaries and the transformer's contextual representations can be more effectively leveraged.

Though the transformer model does a much better job of identifying substances, the drawback is that it takes much longer to train and deploy. Due to the time required to deploy the transformer model on the entire corpus, we chose to deploy it on a random sample of 250,000 comments. Although this is a somewhat small subset of the corpus, it still captures enough entities for meaningful analysis, as there were almost 16,000 comments randomly selected from each subreddit.

Once the NER model had been trained on all of the annotated data, we deployed it across the random sample of 250,000 comments to extract as many substances and effects as possible. At the end of this process, we were left with very large sets of 15,031 substances, representing a broad capture of substances and effects mentioned across the corpus.

2.1.5 FastText Embeddings

Following entity extraction using the custom NER model, vector representations were required to support downstream clustering. To generate these representations, FastText embeddings were trained using the Gensim library with an embedding dimension of 100.

FastText was selected because it represents words as a composition of character n-grams rather than as entire tokens. This subword modeling approach provides three advantages for our corpus. First, it improves representation of rare or sparsely occurring entities, which are common in social media posts. Second, it handles out-of-vocabulary terms more effectively by leveraging shared character patterns. Third, it is well suited for domain-specific terminology, including chemical compounds, slang, abbreviations, and misspellings that frequently appear in the data.

The resulting 100-dimensional FastText embeddings provided dense vector representations of extracted entities, enabling similarity-based clustering in the embedding space.

2.1.6 HDBSCAN

HDBSCAN was applied to the FastText embeddings to cluster semantically similar entities. The objective of this step was to collapse terms within each cluster into a single canonical term, thereby reducing redundancy and standardizing terminology for downstream analysis. Initial clustering results indicated that a substantial proportion of entities were assigned to the noise cluster, reflecting the density-based nature of HDBSCAN and the variability present in social media comments.

To address this, the minimum cluster size was reduced, allowing smaller but still meaningful clusters to form. While soft clustering was considered as a way to reassign noise points based on membership probabilities, it introduced additional noise and reduced cluster interpretability. Instead, a smaller minimum cluster size was chosen as a more controlled approach, with the tradeoff of requiring

additional manual labeling. This decision prioritized cleaner clusters and more reliable canonical term assignments for downstream analysis.

2.1.7 Deduplication of Entities

Following clustering, entities within each cluster were consolidated under a single canonical label. Cluster-level labels were assigned through manual review, and all terms within a cluster were mapped to the selected representative term. This process standardized semantically equivalent or closely related entities to a consistent form. Clusters were also reviewed to ensure semantic coherence. Terms that did not appropriately belong within a cluster were removed prior to final consolidation.

2.3 Sentiment Model Selection & Application Development

2.3.1 Model Selection

Our team conducted research on publicly available sentiment models and the trade-offs between training a custom model and using existing pre-trained models. We prioritized robustness to informal language (e.g., slang and abbreviations) and the ability to understand contextual meaning. We evaluated models using a combination of background research on model characteristics and qualitative assessment, such as comparing generated sentiment scores against post content to determine how accurately sentiment was captured.

Transformer-based models demonstrated strong contextual understanding and the ability to handle slang and informal, user-generated text more effectively, outperforming lexicon-based models while requiring significantly less time and labor than custom-trained models. Based on these advantages, we selected transformer-based models for this analysis.

We selected the Ki-Element Multilingual Valence–Arousal (VA) model and the SamLowe GoEmotions sentiment analysis model to capture complementary aspects of sentiment. The VA model produces continuous valence scores representing sentiment polarity, scaled from –1 to 1 for consistent interpretation, where –1 indicates highly negative sentiment and 1 indicates highly positive sentiment. The GoEmotions model generates discrete emotion labels reflecting specific emotional states and operates as a multi-label classifier, producing probability scores across 28 emotion categories. Due to time and resource constraints, we utilized only the highest-probability label for each input, treating the model as a single-label classifier. After preprocessing, we tokenized the cleaned text and performed model inference for each post. We stored the resulting emotion labels and valence scores alongside user identifiers, subreddit, and timestamps, enabling downstream analyses such as temporal aggregation and longitudinal trend analysis.

We validated model outputs by reviewing sampled predictions against the original text. We then aggregated results at the monthly and substance-category level to support comparative analysis across time and categories. To process the dataset efficiently, we ran inference in batches on a GPU and saved intermediate outputs after each batch to ensure fault tolerance and allow recovery from interruptions.

2.3.2 Application Development

We developed an interactive web application to host the findings of this analysis alongside a substance-effect association network analysis. SENSE (the Substance-Effect Network and Sentiment Analysis tool) is a publicly available web-based Streamlit application with two pages. The Sentiment Explorer displays aggregate trends in user sentiment over time, as well as an interactive heatmap of emotions expressed by users discussing each substance present. Nineteen unique substances are represented in the

Sentiment Explorer, including hallucinogens such as LSD and ketamine, as well as other substances including tobacco products and opioids. SENSE also hosts the Substance-Effect Network. For more information on the Substance-Effect Network, see *Modeling Substance-Effect Relationships in Hallucinogen Use Through Online Conversations*.

3. Results

The longitudinal analysis revealed clear trends in both individual sentiment trajectories and overall sentiment patterns. Using the Multilingual VA model was used to assess temporal trends by analyzing monthly average valence scores, which were visualized using line plots. In parallel, in the GoEmotions model, we identified the dominant emotion for each substance by calculating the proportion of each emotion label relative to the total, excluding neutral classifications.

3.1 Longitudinal Analysis

Across substance terms, most longitudinal sentiment trends remained relatively stable across user-aligned timelines, with no consistent upward or downward trends. Average valence scores for the majority of categories fluctuated but generally hovered around zero, indicating largely neutral sentiment overall. However, not all substances followed this pattern, with some showing more distinct sentiment behavior. For example, opioid-related discussions consistently showed a negative average valence across user posting histories, as shown in Figure 3, suggesting a persistently more negative sentiment associated with these substances. This pattern may reflect the adverse experiences and risks linked to opioid use.

Change Over Tenure in Emotional Valence ③

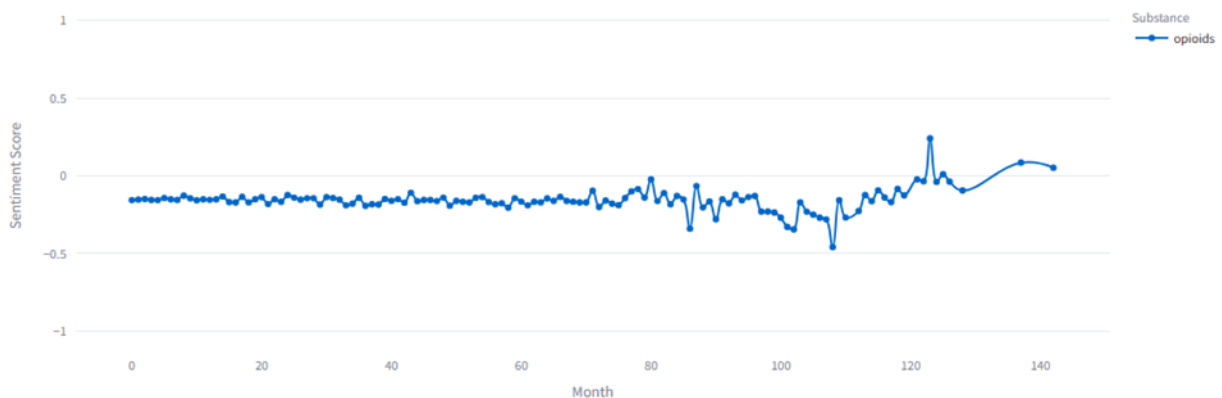


Figure 3: Average Change in Sentiment Over Tenure for Opioids

3.2 GoEmotion Model

GoEmotions is trained on more than 58,000 Reddit comments and classifies text into 27 fine-grained emotion categories (such as admiration, anger, confusion, and joy), as well as a neutral category. It supports multi-label classification, allowing a single comment to express multiple emotions, and captures nuanced emotional expression beyond simple positive or negative sentiment. We removed the neutral label due to its high frequency, which would otherwise have dominated the distribution, and excluded grief because it did not appear as a label in any of our data. As a result, our analysis and heatmap only include 26 emotion labels which are shown in Appendix C.

The heatmap reveals consistent patterns across substances, with approval emerging as the prevalent emotion in most substance terms. Some substances display more distinct emotional profiles; for example, DXM shows a higher proportion of amusement which might be due to DXM effects and how users describe the substance's experience.

3.3 Sentiment Explorer

The second page of our web application, the Sentiment Explorer, allows users to navigate and download the findings from our sentiment analysis. The Sentiment Explorer hosts data on 19 substance terms: amanita, benzodiazepines, cannabis, DMT, DXM, fentanyl, hallucinogens, ketamine, kratom, LSD, MDMA, mescaline, methamphetamine, nicotine vaping, opioids, psilocybin, salvia, stimulants, and tobacco products. The tool visualizes changes in Reddit users' feelings about hallucinogenic substances over user-aligned timelines, as well as emotions identified in posts related to selected substance terms.

The longitudinal sentiment analysis is displayed in a line chart where users can select substances to view. The visualization displays the average change in sentiment score by month since each user's first observed post for the selected substance. Users can filter by substance or trend direction. We classify trend, defined as the average change in sentiment over this user-aligned timeline, as positive, negative, or neutral using the same thresholds as the sentiment score. Figure 4 displays the average change in sentiment over user-aligned time for fentanyl and psilocybin.

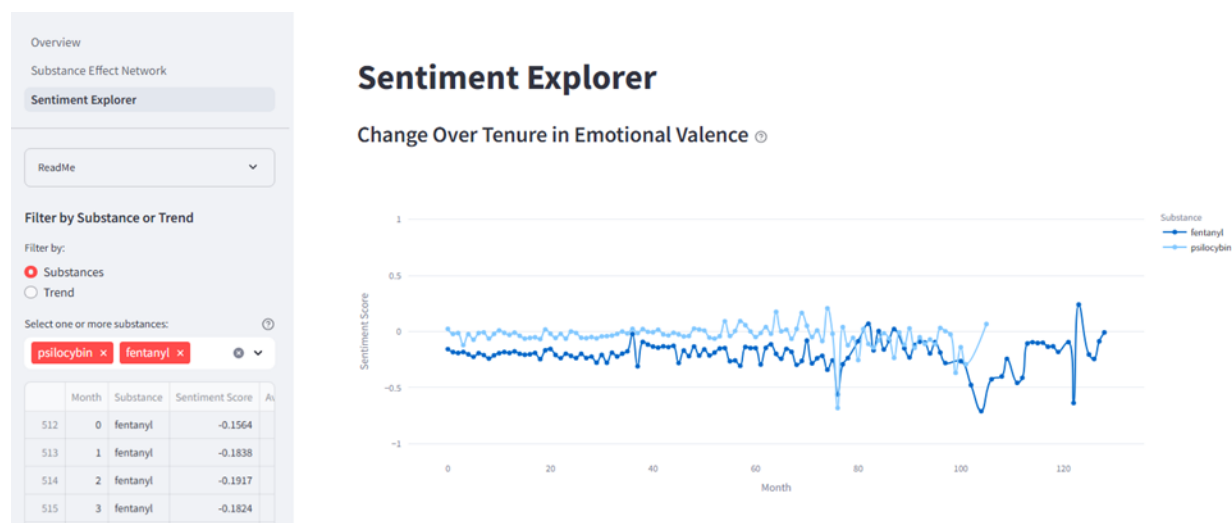


Figure 4. Sentiment Change Over Tenure by Substance, Filtered by Fentanyl and Psilocybin.

This visualization is useful for comparing changes in average valence between substance terms. At most time points, the average sentiment score for psilocybin is more positive than that of fentanyl. Hovering over a time point in the graph displays the substance, month, that substance's average sentiment score for that month, and the total number of comments represented. In the data preview in the sidebar, users can view and download the average sentiment score, total average valence trend, trend direction, and comment count for each month per substance selected.

Emotion distribution by substance term is visualized using a heatmap. Darker tones indicate a greater percentage of posts in our data that indicate the identified emotion for the identified substance. Hovering over a cell in the heatmap allows the user to see the count of posts with that emotion, as well as the percent of posts discussing the identified substances for which that emotion is present. After

excluding “neutral” from this visualization, there were 26 emotions present in the heatmap. Figure 5 displays the heatmap with the cursor hovering over psilocybin and confusion to show the count and percent of posts discussing psilocybin that are identified with that emotion.

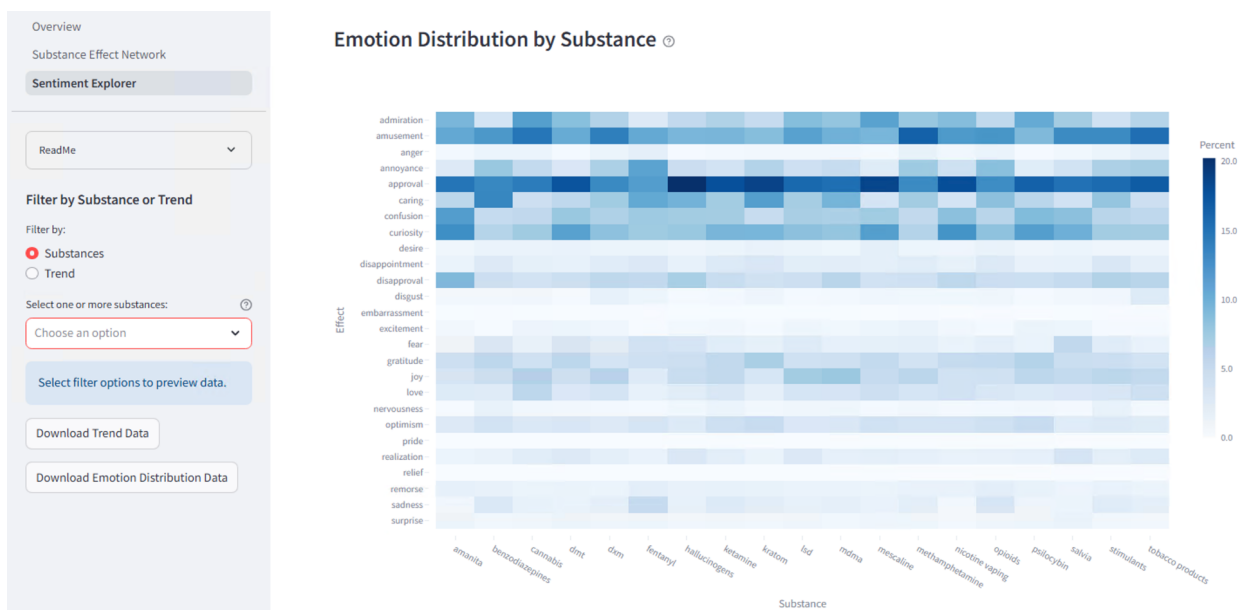


Figure 5. Sentiment Explorer Emotion Distribution by Substance

In our dataset, 13,571 psilocybin related comments (9.0% of all comments about psilocybin) express confusion. The Sentiment Explorer complements the Substance-Effect Network to provide users with a comprehensive view of how Reddit users discuss hallucinogenic substances. Data from each page can be downloaded and analyzed externally, a feature our team believes will support future research on these substances and act as a foundation for broader understanding of the ways people experience these substances in a non-clinical environment.

4. Discussion

4.1 Principal Findings

This study shows that large-scale social media data can reveal meaningful patterns in sentiment related to substances. Across the 19 substance terms analyzed, longitudinal sentiment trends remained relatively stable, with most average valence scores clustering near neutral. For example, psilocybin-related discussions generally remained neutral over time. Despite this general pattern, some substances do display a different pattern. For example, fentanyl consistently exhibited negative sentiment over time with a valence score regularly below zero. Emotion classification revealed consistent patterns, with approval as the most prevalent emotion. Some substances showed slight deviations from this pattern such as DXM which exhibited higher proportions of amusement which might be due to DXM effects and how users describe the substance’s experience.

4.2 Contributions

The Sentiment Explorer, alongside the Substance Effect Network in the SENSE application, provides a strong foundation for future research on hallucinogen use behavior and sentiment.

4.2.1 Understanding User Behavior

Understanding how frequently individuals discuss and engage with hallucinogenic substances can provide researchers and public health officials with a more comprehensive view of usage patterns. In this study, we use frequency of posting within substance-specific subreddits as a proxy for frequency of use, under the assumption that increased engagement in these forums reflects greater involvement with the substance. Accordingly, substances associated with higher posting frequencies may indicate more frequent use among individuals in our sample.

4.2.2 Identifying Temporal Change Patterns in Perception

As users repeatedly take hallucinogens, their perception toward those substances may change as their physiological tolerance and comfort with the substance changes. Identifying positive or negative trends with repeated use may allow researchers to better understand which substances may be related to more positive or more negative experiences over time (Volkow et. al., 2011). This may inform strategies related to therapeutic and clinical use of these substances for. It may further clarify how concerns such as addiction and decreased sensitivity relate to use behavior and frequency.

4.2.3 Informing Harm Reduction Initiatives

Understanding the feelings that users express related to specific substances may allow policymakers to target harm reduction initiatives toward substances most likely to induce a negative response from users. For example, in the Emotion Distribution by Substance Heatmap, there is a greater percent of comments expressing disapproval for amanita compared to any other substance. Amanita muscaria is often ingested inadvertently after being misidentified as other hallucinogenic mushrooms such as psilocybin (Rampolli et. al., 2021). Ingestion of amanita often causes more negative effects compared to these substances, including respiratory failure and stomach pain. Identifying other such trends toward negative sentiment in our tool could inform awareness and other harm reduction programs to help users avoid accidental ingestion.

4.4 Limitations

Due to the nature of social media conversations and sentiment analysis models, this project has a few key limitations. First, we had to assume that comments expressed truthful and firsthand sentiment toward these substances. However, this is unlikely to be the case with every comment. For example, a comment identified as having a negative valence toward LSD might have been discussing the user's own experience, but it also could have been discussing negative feelings toward the substance even though the commenter never used it personally. We assumed that the proportion of comments that were not firsthand, truthful accounts was small enough that it did not affect the aggregate sentiment and emotions identified in the Sentiment Explorer.

Second, social media conversations often include sarcasm, slang, or other internet-specific speech patterns that are not fully represented by the GoEmotion modeling process. While we did select the GoEmotion model because it handles these forms of text better than the alternatives, language evolves so quickly on the internet that we cannot guarantee that it successfully captured all of the nuances in our data.

Third, the GoEmotion model is limited to identifying the emotions expressed within a single post. Therefore, if a commenter's emotion is expressed across multiple sentences, the model does not adequately capture it. Because the Sentiment Explorer only displays the top emotion for each post, it once again does not fully capture all nuances in the text.

Additionally, we cannot definitively say that the emotions expressed related to a substance are causal. Changes in sentiment over time may coincide with substance use, but they may also be influenced by external factors such as personal circumstances, co-occurring substance use, or broader social and environmental dynamics. The analysis does not determine whether a substance contributes to a particular emotional state or is discussed in response to it, and should not be used to infer directionality.

Because some users comment more frequently about a substance, their sentiment and opinion on that substance is over-represented in our analysis. This may potentially introduce bias into our trend estimates. These users may differ systematically from the broader population in terms of experience, engagement, or attitudes toward substance use. Therefore, observed temporal trends may reflect the behavior and perspectives of a highly engaged subset of users rather than generalizable patterns across all users.

Finally, the emotion classification model itself introduces limitations. Pre-trained models such as GoEmotions are developed on general-domain text and may not fully capture the nuances of substance-related discussions, including slang, irony, or domain-specific expressions of affect. This can lead to misclassification or oversimplification of emotional states. Further, human emotion and sentiment are inherently complex and cannot be fully understood by a model without reading the whole comment and understanding its broader context.

Given these limitations, the results from our analysis and the Sentiment Explorer should be used to assess broad trends in sentiment, rather than as concrete, reproducible changes in sentiment. Despite the limitations, this approach empowers further exploration into how drug users feel about the substances they take, an important step to better understanding drug culture and habits as a whole.

5. Conclusion

The use of hallucinogenic substances is widely discussed in online social media forums such as Reddit. These conversations offer an opportunity to better understand patterns of drug use behavior and perception among a large community of individual users. The longitudinal sentiment and emotion distribution analyses demonstrate the potential to extract valuable insights from social media conversations, especially related to sensitive topics such as illicit substance use.

Identifying the positive or negative trend in sentiment over time for the users of each substance can help users better understand the potential risks of extended use. Although the average trends for each substance in our corpus were relatively small, our methodology may be replicated on a more comprehensive collection of text in the future to create a more robust sense of the change in user feelings over time.

We identified patterns in emotions expressed by users when discussing select substances, as well as trends in user sentiment over times of prolonged use. We found that the most frequent emotion expressed for every substance in our data was approval. After approval, frequently expressed emotions ranged from curiosity and amusement to confusion and annoyance. Understanding the most commonly expressed emotion for each substance can inform future harm reduction research.

The publication of our findings in the Sentiment Explorer page of the SENSE application empowers our target audience, including public health officials, researchers, clinicians, and drug users, to better

compare user perception of these substances and facilitate a deeper understanding of how hallucinogens are discussed online.

6. References

- Rampolli, F. I., Kamler, P., Carnevale Carlino, C., & Bedussi, F. (2021). The Deceptive Mushroom: Accidental Amanita Muscaria Poisoning. *European Journal of Case Reports in Internal Medicine*, 8(2). https://doi.org/10.12890/2021_002212
- Volkow, N. D., Wang, G., Fowler, J. S., Tomasi, D., Telang, F., & Baler, R. (2010). Addiction: Decreased reward sensitivity and increased expectation sensitivity conspire to overwhelm the brain's control circuit. *BioEssays*, 32(9), 748–755. <https://doi.org/10.1002/bies.201000042>

7. Appendix

Appendix A: Subreddits in Corpus

Subreddit	Description	Comment Count	Date Created
r/Ayahuasca	Information, discussions and personal experiences about the shamanistic plant medicine Ayahuasca.	176,570	Dec 26, 2010
r/DMT	A community connected by N,N-Dimethyltryptamine.	1,380,311	Apr 28, 2008
r/dxm	A community for discussion about responsible use of the powerful dissociative cough suppressant - Dextromethorphan! Harm Reduction Drug Culture 18+	1,008,692	Sep 16, 2009
r/ketamine	A subreddit for discussing the recreational use of ketamine with a focus on harm reduction.	334,703	Jul 20, 2011
r/LSD	A kind, open-minded community dedicated to Lysergic Acid Diethylamide-25.	3,946,970	Apr 28, 2008
r/LSDTripLifeHacks	The name is self explanatory...	72,788	Oct 3, 2013
r/MDMA	A harm reduction community for the responsible use of MDMA, otherwise known as molly, mandy, ecstasy, etc.	1,370,696	Apr 18, 2009
r/microdosing	This is a community for discussion pertaining to microdosing research, experiments, regimens and experiences. The most probable candidates for microdosing are psychedelics, but we encourage dialogue on the effects of any drugs at sub-threshold dosage.	447,874	Oct 16, 2013
r/opiates	Discussion of all things related to the narcotics known as opiates, from harm-reduction to pharmacology.	4,136,050	Jun 24, 2009
r/psilocybin	Exploring the Mysteries of Nature's Own Mind Melder Welcome to r/Psilocybin! 🍄 The Most active Magic Mushroom Community on Reddit! 🍄 This community explores psilocybin, a fascinating hallucinogen with historic significance and potential therapeutic use. We discuss its scientific effects, potential in mental health, and	47,928	Aug 5, 2012

	its global history. Upholding Reddit's rules, we advocate for responsible, informed dialogue. Join our journey into the captivating world of psilocybin and magic mushrooms! 🍄🍄		
r/Psychedelics	r/psychedelics is a subreddit for people who are interested in exploring the effects and potential benefits of psychedelics. Whether you are a novice or an experienced psychonaut, you can find informative posts, personal stories, memes, and discussions about various aspects of psychedelics. As well as discuss psychedelic history, culture, science, art, spirituality, and harm reduction. 🍄🌈👁️	611,577	Jan 12, 2009
r/Psychnaut	A Psychonaut is a person who explores activities by which altered states of consciousness are induced and utilized for spiritual purposes or the exploration of the human condition, including shamanism, sensory deprivation, and both archaic and modern users of entheogenic substances, in order to gain deeper insights into the mind and spirituality.	2,068,192	Dec 8, 2008
r/Salvia	A place to discuss Salvia divinorum, a psychoactive plant containing the potent psychedelic Salvinorin A, used traditionally by Mazatec shamans in shamanic and spiritual rituals. The safe and responsible use of this plant is highly encouraged, as Salvinorin A is the most powerful naturally-found hallucinogen.	174,569	Apr 2, 2009
r/shrooms	A place to chat about the growing, foraging and experience of magic mushrooms. Mostly about mushrooms with psilocybin, but all mind-altering fungi are welcome.	3,443,778	Feb 11, 2009
r/TherapeuticKetamine	A place for patients and healthcare providers to discuss the use of prescription ketamine treatments.	135,095	Oct 9, 2017
r/weed	The subreddit for all things weed! Talk strains, first times, positive/negative experiences. Share your photos and videos of sexy buds, plants, or cherished pieces.	4,511,872	Mar 30, 2008

Appendix B: Prompt for Gemini Annotation

"

You are an expert NER annotator for a dataset about psychoactive substances and their effects, drawn from Reddit communities (r/psychedelics, r/LSD, r/Drugs, r/shrooms, r/MDMA, etc.).

Your task: identify every mention of a SUBSTANCE or EFFECT in the comment.

Return the exact span text as it appears — including misspellings and slang.

ENTITY TYPES

SUBSTANCE — any psychoactive, mind-altering, or recreational substance.

Include ALL of the following:

- Psychedelics / classic hallucinogens:

LSD → acid, lucy, L, tabs, tab, blotter, hit, dose, lysergic

Psilocybin mushrooms → shrooms, mushies, shroomies, melmac, golden teacher,
penis envy, cubes, cubensis, magic mushrooms, boomers, fungi

DMT → deemster, the spirit molecule, dimitri, spice

Ayahuasca → aya, vine, yage

Mescaline → peyote, mesc, san pedro

2C-B, 2C-I, 2C-x, NBOMe, 25i, 25c, RC (research chemicals)

4-AcO-DMT, 4-HO-MET, 5-MeO-DMT, etc.

- Empathogens / entactogens:

MDMA → molly, ecstasy, mandy, rolls, X, XTC, rollies, thizz, E

MDA → sally

- Dissociatives:

Ketamine → K, ket, special K, vitamin K

PCP → angel dust

DXM → dex, dextromethorphan (found in Robitussin, Mucinex DM)

Nitrous oxide → nitrous, nos, whippets, laughing gas, balloons

- Cannabis & cannabinoids:

weed, pot, marijuana, bud, ganja, herb, dank, reefer, Mary Jane, trees,

joints (js), blunts, spliff, hash, kief, rosin, dabs, concentrate,

THC, CBD, flower, cart, cartridge

- Stimulants:

cocaine → coke, blow, snow, charlie, nose candy

amphetamine → speed, uppers, whiz

methamphetamine → meth, crystal, ice, glass, tina

prescription stimulants: adderall, vyvanse, ritalin

- Depressants / sedatives:

alcohol → booze, beer, liquor, drinks, shots, ethanol

benzodiazepines → benzos, xanax, klonopin, valium, ativan

GHB / GBL, opioids → heroin, fentanyl, oxy, morphine, codeine

- Other pharmaceuticals used recreationally:

Mucinex (DXM source), Robitussin, gabapentin, pregabalin, tramadol

- Strain / cultivar names that clearly refer to a specific substance

(e.g., "melmac", "golden teacher", "penis envy", "gorilla glue", "OG kush")

- Include intentional misspellings: "musrooms", "shrroms", "shroomie", "acud", "mdmd", etc.

- This list is NOT exhaustive. Use your knowledge of drug culture, Reddit slang, and context to identify any substance not listed here. If the surrounding text strongly implies something is a substance or effect (e.g. an unusual word used in a dosing, trip report, or harm reduction context), label it.

EFFECT — any physical, psychological, perceptual, or emotional effect of substance use.

- Perceptual / visual:

visuals, OEVs (open-eye visuals), CEVs (closed-eye visuals),
tracers, fractals, patterns, hallucinations, geometric patterns,
visual distortions, halos, auras, color enhancement, things breathing

- Psychological / cognitive:

ego death, ego dissolution, ego loss, depersonalization, derealization,
thought loops, mind racing, confusion, dissociation, psychosis,
introspection, anxiety, paranoia, panic, fear, overwhelm

- Emotional:

euphoria, empathy, love, warmth, bliss, sadness, emotional intensity,
emotional release, mood lift, depression (post-use)

- Physical / somatic:

nausea, body load, jaw clenching, bruxism, tachycardia, heart racing,
sweating, dizziness, vomiting, headache, muscle tension, vasoconstriction,
body high, tingling, numbness, stimulation, sedation, fatigue

- Auditory / other sensory:

music enhancement, auditory hallucinations, sound distortions, synesthesia

- Temporal / state:

time dilation, time distortion, peak, comedown, afterglow, crash,
trip (when describing the experience/state), high, roll, buzz,
k-hole, space, HPPD, flashback

- Sleep-related (when attributed to substance use):

vivid dreams, night terrors, insomnia, disrupted sleep

- Use context to identify any effect not listed here. Unusual descriptors used in the

context of a trip or drug experience should be labeled.

RULES

1. Extract the MINIMAL span that names the entity. Do not include surrounding words.

✓ "shrooms" X "some shrooms"

2. Include EVERY occurrence, even duplicates in the same comment.

3. Preserve the EXACT original spelling (typos, lowercase, abbreviations).

4. When uncertain, INCLUDE the entity — over-labeling is preferable for NER training.

5. A comment may have ZERO entities — return an empty list.

6. DO NOT label:

- Paraphernalia / equipment alone (bong, pipe, gauze, rolling paper, syringe)
- Dosage amounts alone ("400ug", "3.5g", "800mg") — label the substance if named nearby
- General idioms unrelated to drug use ("that's trippy" if no substance context)

FEW-SHOT EXAMPLES

--- Example 1 ---

Comment ID: 101

Text: "you can have really vivid dreams and night terrors the following week or two after mdma. drink plenty of water and get some sun. also highly recommend not using mdma until you're at least 18."

Output:


```
{"id": 101, "text": "you can have really vivid dreams and night terrors the following week or two after mdma. drink plenty of water and get some sun. also highly recommend not using mdma until you're at least 18.", "entities": [{"span": "vivid dreams", "label": "EFFECT"}, {"span": "night terrors", "label": "EFFECT"}, {"span": "mdma", "label": "SUBSTANCE"}]}
```

--- Example 2 ---

Comment ID: 102

Text: "i probably wouldn't bother with the mucinex - but it probably won't do much harm (just increase your nausea). either way, have fun :)"

Output:

```
{"id": 102, "text": "i probably wouldn't bother with the mucinex - but it probably won't do much harm (just increase your nausea). either way, have fun :)", "entities": [{"span": "mucinex", "label": "SUBSTANCE"}, {"span": "nausea", "label": "EFFECT"}]}
```

--- Example 3 ---

Comment ID: 103

Text: "uh no. chances are they're probably in the 100ug-120ug range and you do not wanna do 800ug for a first trip. i would do 2 max"

Output:

```
{"id": 103, "text": "uh no. chances are they're probably in the 100ug-120ug range and you do not wanna do 800ug for a first trip. i would do 2 max", "entities": [{"span": "trip", "label": "EFFECT"}]}
```

--- Example 4 ---

Comment ID: 104

Text: "i had a 3.5g of melmac and it looked exactly like those"

Output:

```
{"id": 104, "text": "i had a 3.5g of melmac and it looked exactly like those", "entities": [{"span": "melmac", "label": "SUBSTANCE"}]}
```

--- Example 5 ---

Comment ID: 105

Text: "mixing ket and molly gave me insane CEVs and I k-holed for like 20 mins, then the jaw clenching started"

Output:

```
{"id": 105, "text": "mixing ket and molly gave me insane CEVs and I k-holed for like 20 mins, then the jaw clenching started", "entities": [{"span": "ket", "label": "SUBSTANCE"}, {"span": "molly", "label": "SUBSTANCE"}, {"span": "CEVs", "label": "EFFECT"}, {"span": "k-holed", "label": "EFFECT"}, {"span": "jaw clenching", "label": "EFFECT"}]}
```

--- Example 6 ---

Comment ID: 106

Text: "are you running low?"

Output:

```
{"id": 106, "text": "are you running low?", "entities": []}
```

--- Example 7 ---

Comment ID: 107

Text: "shroomie puns"

Output:

```
{"id": 107, "text": "shroomie puns", "entities": [{"span": "shroomie", "label": "SUBSTANCE"}]}
```

Return ONLY a single valid JSON object. No markdown, no explanation.

"

Appendix C: SamLowe Emotion Labels

Emotion Labels
Admiration
Amusement
Anger
Annoyance
Approval
Caring
Confusing
Curiosity
Desire
Disappointment
Disapproval
Disgust
Embarrassment
Excitement
Fear
Gratitude
Joy
Love
Nervousness
Optimism
Pride
Realization
Relief
Remorse
Sadness
Surprise